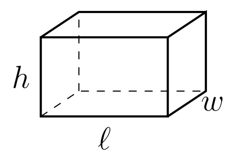
Surface Area and Scale Design Workshop

Focus: Prism surface area, cylinder surface area, composite plan areas, and scale-factor design.

Directions: Every problem takes more than one step. Write the formula you are using, substitute the given measurements, and label your answer with its unit. Use $\pi \approx 3.14159$ and round money-free area answers to two decimal places only when the answer is not exact. Diagrams show the shape and the letters used for its measurements; the numbers are in the problem.

1. A closed cardboard carton is a rectangular prism measuring $\ell = 12$ cm long, $w = 8$ cm wide, and $h = 5$ cm high.



Find the total surface area of the carton — the cardboard needed for all six faces.

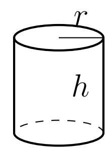
Answer: ________________ cm^2

2. A floor plan is drawn to the scale 1 cm on the drawing = 4 m in the building. On the drawing, a rectangular room measures 6.5 cm by 3.25 cm.

Find the **actual perimeter** of the room — the length of skirting board needed to run right around it.

Answer: ________________ m

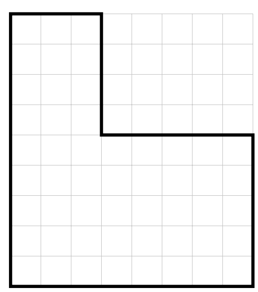
3. A closed tin can is a cylinder with radius $r = 3$ cm and height $h = 8$ cm.



Find the total surface area of the can: the curved side plus both circular ends.

Answer: ________________ cm^2

4. The plan below shows the floor of a workshop. Each grid square is 1 metre by 1 metre.



Split the plan into rectangles and find the total floor area.

Answer: ________________ m^2

5. A garden shed is a rectangular prism 6 m long, 4 m wide and 3 m high. The four walls and the flat roof will be painted; the concrete floor will **not** be painted.

Find the area to be painted.

Answer: ________________ m^2

6. On the same 1 cm = 4 m drawing, a rectangular deck measures 5 cm by 3 cm.

Find the **actual area** of the deck. (Careful: the scale factor for lengths is not the scale factor for areas.)

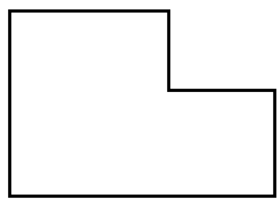
Answer: ________________ m^2

7. A paper label is wrapped around the curved side of a cylindrical can of radius 4 cm and height 11 cm. The label is glued with an overlap, so its length is 1 cm longer than the circumference of the can, and it is as tall as the can.

Find the area of the paper label.

Answer: ________________ cm^2

8. A shed design uses two identical L-shaped side panels, drawn below. Each panel is a 10 m by 4 m rectangle with a 6 m by 3 m rectangle attached along the top of its left-hand side.



Find the total area of material needed for **both** panels.

Answer: ________________ m^2

9. A designer builds a model of a shipping crate at the scale 1 : 5, so every length on the real crate is 5 times the matching length on the model. The model's total surface area is 148 cm^2.

Find the total surface area of the real crate, in square centimetres.

Answer: ________________ cm^2

10. A closed box has a square base measuring 8 cm by 8 cm, and its total surface area is 448 cm^2.

Work backwards from the surface-area formula to find the height of the box.

Answer: ________________ cm

11. **Design decision.** Two closed boxes hold the same volume of 216 cubic centimetres. Box A is a cube measuring 6 cm by 6 cm by 6 cm. Box B measures 12 cm by 6 cm by 3 cm.

Which box uses less cardboard? Support your answer with a full surface-area comparison for both boxes, then give one practical reason a designer might still choose the other box.

12. **Synthesis.** A scale drawing of a cylindrical water tank uses the scale 1 cm = 2.5 m. On the drawing the tank is 1.6 cm across (its diameter) and 3.2 cm tall. The real tank will have its curved wall and its flat circular top painted; the base sits on the ground and is not painted.

Find the actual area to be painted.

Answer: ________________ m^2